Supplementary Material

S1. Derivation of transcription and translation kinetic rate expressions

We model transcription as a four-step process based on the classical work of McClure (1) and as developed previously (2). RNA polymerase (RNAP) binds the promoter of gene j to form a closed complex, which isomerizes to an open complex, from which either abortive initiation or productive elongation occurs:

$$\mathcal{G}_j + R_X \rightleftharpoons (\mathcal{G}_j : R_X)_C \tag{S1}$$
$$(\mathcal{G}_j : R_X)_C \rightarrow (\mathcal{G}_j : R_X)_O \tag{S2}$$
$$(\mathcal{G}_j : R_X)_O \rightarrow R_X + \mathcal{G}_j \tag{S3}$$
$$(\mathcal{G}_j : R_X)_O \rightarrow m_j + R_X + \mathcal{G}_j \tag{S4}$$

where $\mathcal{G}_j$ denotes the concentration of gene j, R_X is the concentration of *free* RNAP, $(\mathcal{G}_j : R_X)_C$ and $(\mathcal{G}_j : R_X)_O$ are the closed and open complex concentrations, and m_j is the mRNA product. The key idea is that RNAP acts as a pseudo-enzyme: it binds its substrate (the gene), processes it, and dissociates, analogous to Michaelis-Menten kinetics.

Let k_+ (conc^{-1} t^{-1}) and k_- (t^{-1}) denote the on/off rate constants for RNAP binding at the promoter, k_I (t^{-1}) the rate constant for open complex formation, k_A (t^{-1}) the abortive initiation rate constant, and $k_{E,j}^X$ (t^{-1}) the elongation rate constant for gene j. The material balances for the closed and open complexes are:

$$\frac{d}{dt}(\mathcal{G}_j : R_X)_C = k_+ \, \mathcal{G}_j \, R_X - k_- \, (\mathcal{G}_j : R_X)_C - k_I \, (\mathcal{G}_j : R_X)_C \tag{S5}$$
$$\frac{d}{dt}(\mathcal{G}_j : R_X)_O = k_I \, (\mathcal{G}_j : R_X)_C - k_A \, (\mathcal{G}_j : R_X)_O - k_{E,j}^X \, (\mathcal{G}_j : R_X)_O \tag{S6}$$

Invoking the pseudo-steady-state approximation (justified because complex formation and dissociation are fast compared to mRNA accumulation), we set Eqns. S5–S6 to zero and solve for the steady-state complex concentrations:

$$(\mathcal{G}_j : R_X)_C = \left(\frac{k_+}{k_- + k_I}\right) \mathcal{G}_j \, R_X \equiv K_{X,j}^{-1} \, \mathcal{G}_j \, R_X \tag{S7}$$
$$(\mathcal{G}_j : R_X)_O = \left(\frac{k_I}{k_A + k_{E,j}^X}\right) (\mathcal{G}_j : R_X)_C \equiv \tau_{X,j}^{-1} \, (\mathcal{G}_j : R_X)_C \tag{S8}$$

where the saturation constant $K_{X,j}^{-1} \equiv k_+/(k_- + k_I)$ has units of inverse concentration and governs the affinity of RNAP for the promoter, and the dimensionless time constant $\tau_{X,j}^{-1} \equiv k_I/(k_A + k_{E,j}^X)$ compares the rate of open complex formation to the combined rates of elongation and abortive initiation. Combining these expressions, the open complex concentration is $(\mathcal{G}_j : R_X)_O = K_{X,j}^{-1} \, \tau_{X,j}^{-1} \, \mathcal{G}_j \, R_X$, and the rate of mRNA production for gene j is proportional to this quantity:

$$\boxed{r_{X,j} = k_{E,j}^X \, (\mathcal{G}_j : R_X)_O = k_{E,j}^X \, K_{X,j}^{-1} \, \tau_{X,j}^{-1} \, \mathcal{G}_j \, R_X} \tag{S9}$$

where R_X is the free (unbound) RNAP concentration. The elongation rate constant $k_{E,j}^X$ is related to the polymerase elongation speed $\dot{v}_X$ (nt/s) and the gene coding length $l_{G,j}$ (nt) by $k_{E,j}^X = \dot{v}_X / l_{G,j}$, and the time constant is parameterized as $\tau_{X,j} = (k_{E,j}^X / k_I) \cdot \hat{\tau}_{X,j}$, where $k_I = 1/t_{\text{init},X}$ is the initiation rate constant estimated from McClure assays (1) and $\hat{\tau}_{X,j}$ is a dimensionless gene-specific modifier estimated from data.

An identical derivation applies to translation, with ribosome (R_L) replacing RNAP and mRNA (m_j) replacing gene ($\mathcal{G}_j$):

$$\boxed{r_{L,j} = k_{E,j}^L \, K_{L,j}^{-1} \, \tau_{L,j}^{-1} \, m_j \, R_L} \tag{S10}$$

where $k_{E,j}^L = \dot{v}_L / l_{P,j}$ is the translation elongation rate constant, $l_{P,j}$ is the protein length in amino acids, $K_{L,j}$ is the translation saturation constant, and $\tau_{L,j} = (k_{E,j}^L / k_{I,L}) \, \hat{\tau}_{L,j}$ is the translation time constant.

The time constant $\tau_{X,j}$ governs whether gene j is elongation-limited or initiation-limited. Assuming abortive initiation is slow ($k_A \ll k_I, k_{E,j}^X$), then $\tau_{X,j} \simeq k_{E,j}^X / k_I$. In the initiation-limited regime ($\tau_{X,j} \gg 1$), the transcription rate reduces to the familiar Michaelis-Menten form $r_{X,j} \simeq (k_{E,j}^X / \tau_{X,j}) \, R_{X,T} \, \mathcal{G}_j / (K_{X,j} + \mathcal{G}_j)$, while in the elongation-limited regime ($\tau_{X,j} \ll 1$), the effective saturation constant is reduced by $\tau_{X,j}$, reflecting tighter RNAP binding when elongation is slow.

S2. Free machinery concentrations and competitive allocation

Consider $\mathcal{N}$ genes competing for a shared pool of RNAP with total concentration $R_{X,T}$. At any time, RNAP is either free or sequestered in closed or open complexes:

$$R_{X,T} = R_X + \sum_{j=1}^{\mathcal{N}} \left[(\mathcal{G}_j : R_X)_C + (\mathcal{G}_j : R_X)_O \right] \tag{S11}$$

Substituting the steady-state expressions (Eqns. S7–S8) and factoring out R_X:

$$R_{X,T} = R_X \left[1 + \sum_{j=1}^{\mathcal{N}} \frac{(1 + \tau_{X,j}^{-1}) \, \mathcal{G}_j}{K_{X,j}} \right] \tag{S12}$$

Solving for the free RNAP concentration yields:

$$\boxed{R_X = \frac{R_{X,T}}{1 + \displaystyle\sum_{j=1}^{\mathcal{N}} \frac{(\tau_{X,j} + 1) \, \mathcal{G}_j}{\tau_{X,j} \, K_{X,j}}}} \tag{S13}$$

Each gene j sequesters RNAP in proportion to its concentration $\mathcal{G}_j$, its affinity $K_{X,j}^{-1}$, and the factor $(\tau_{X,j} + 1)/\tau_{X,j}$ which accounts for RNAP trapped in both closed and open complexes. By identical reasoning, the free ribosome concentration is:

$$\boxed{R_L = \frac{R_{L,T}}{1 + \displaystyle\sum_{j=1}^{\mathcal{N}} \frac{(\tau_{L,j} + 1) \, m_j}{\tau_{L,j} \, K_{L,j}}}} \tag{S14}$$

where m_j is the mRNA concentration of gene j, which varies dynamically. Thus R_X is constant in time (since gene concentrations are fixed for linear DNA), while R_L changes as mRNA levels evolve.

For a single gene, substituting Eqn. S13 into Eqn. S9 gives $R_X = R_{X,T}\,\tau_{X,j}\,K_{X,j}/[\tau_{X,j}\,K_{X,j} + (\tau_{X,j}+1)\,\mathcal{G}_j]$. After substitution and cancellation of $\tau_{X,j}\,K_{X,j}$, the transcription rate simplifies to:

$$r_{X,j} = V_{X,j}^{\max}\left(\frac{\mathcal{G}_j}{\tau_{X,j}\,K_{X,j} + (1+\tau_{X,j})\,\mathcal{G}_j}\right) \tag{S15}$$

where $V_{X,j}^{\max} \equiv k_{E,j}^{X}\,R_{X,T} = R_{X,T}\,\dot{v}_X/l_{G,j}$ is the maximum transcription rate. This is the form presented in the main text.

For multiple genes, we define the allocation fraction $f_{X,j} \equiv \mathcal{G}_j/[\tau_{X,j}\,K_{X,j} + (1+\tau_{X,j})\,\mathcal{G}_j]$ and write the free RNAP as $R_{X,\text{free}} = R_{X,T}/(1+\sum_j f_{X,j})$. The transcription rate for gene j in the multi-gene system is then:

$$r_{X,j} = R_{X,\text{free}} \cdot k_{E,j}^{X} \cdot f_{X,j} \tag{S16}$$

or equivalently, $r_{X,j} = V_{X,j}^{\max}\,f_{X,j}$ where now $V_{X,j}^{\max} = R_{X,\text{free}}\,(\dot{v}_X/l_{G,j})$ uses the *free* RNAP concentration rather than the total. This formulation treats RNAP as a shared enzyme that is competitively allocated among genes, analogous to competitive substrate inhibition in enzyme kinetics. The factored form separates the global competition effect (through $R_{X,\text{free}}$, which decreases as more genes sequester RNAP) from the individual gene's kinetics (through $f_{X,j}$, which depends only on gene j's concentration and kinetic parameters).

S3. Statistical thermodynamic derivation of the transcription control function

We model transcriptional regulation using the statistical thermodynamic framework developed by Ackers et al. (3) and adapted for synthetic circuits by Moon et al. (4). The central idea is that the promoter region of gene j can exist in a finite number of discrete microscopic configurations (microstates), each corresponding to a different combination of bound transcription factors and RNA polymerase. At thermal equilibrium, the probability of finding the promoter in microstate s is given by the Boltzmann distribution:

$$p_s = \frac{W_s\,g_s(\mathbf{P})}{\displaystyle\sum_{s' \in \mathcal{C}_j} W_{s'}\,g_{s'}(\mathbf{P})} \tag{S17}$$

where the sum is over all possible configurations $\mathcal{C}_j$ of promoter j. Here $W_s = \exp(-\epsilon_s/k_BT)$ is the Boltzmann weight for configuration s, determined by the pseudo-energy ϵ_s relative to a reference state (the empty promoter, $\epsilon_0 = 0$, $W_0 = 1$), and $g_s(\mathbf{P})$ is a binding function that depends on the concentrations of the relevant transcriptional regulators $\mathbf{P}$. For configurations in which a regulator is bound, g_s includes a Hill-type occupancy term; for the empty or RNAP-only states, $g_s = 1$. A strongly negative ϵ_s (large W_s) indicates a thermodynamically favorable binding interaction, while a positive ϵ_s indicates an unfavorable configuration.

The transcription control function $\bar{u}_j$ is defined as the probability that the promoter is in a configuration that leads to productive transcription:

$$\boxed{\bar{u}_j = \frac{\sum_{s \in \chi_j} W_s \, g_s(\mathbf{P})}{\sum_{s \in \mathcal{C}_j} W_s \, g_s(\mathbf{P})}} \tag{S18}$$

where $\chi_j \subseteq \mathcal{C}_j$ is the subset of transcriptionally active configurations. The value $\bar{u}_j \in [0, 1]$ represents the fraction of time the promoter spends in a productive state and directly modulates the transcription rate.

The gluconate sensor circuit contains three genes, each with a distinct promoter architecture. The P70 promoter driving GntR can exist in three states: (i) empty, (ii) core RNAP bound (without σ_{70}), or (iii) holoenzyme (σ_{70}-RNAP) bound. Both RNAP-bound states are productive, giving:

$$\bar{u}_{\mathrm{GntR}} = \frac{W_{\mathrm{GntR,cRNAP}} + W_{\mathrm{GntR},\sigma_{70}} \, b_{\mathrm{GntR},\sigma_{70}}}{1 + W_{\mathrm{GntR,cRNAP}} + W_{\mathrm{GntR},\sigma_{70}} \, b_{\mathrm{GntR},\sigma_{70}}} \tag{S19}$$

where $W_{\mathrm{GntR,cRNAP}} = \exp(-\epsilon_{\mathrm{GntR,cRNAP}}/k_B T)$ and $W_{\mathrm{GntR},\sigma_{70}} = \exp(-\epsilon_{\mathrm{GntR},\sigma_{70}}/k_B T)$ are Boltzmann weights for core RNAP and holoenzyme binding, respectively, and $b_{\mathrm{GntR},\sigma_{70}} = [\sigma_{70}]^n/(K^n + [\sigma_{70}]^n)$ is a Hill function describing the fractional occupancy of σ_{70} on the holoenzyme.

The modified P70 promoter (mP70) driving Venus includes a GntR operator between the -35 and -10 regions, adding a fourth state in which the GntR repressor is bound to the operator. This state is *not* productive, so GntR occupancy reduces $\bar{u}_{\mathrm{Venus}}$:

$$\bar{u}_{\mathrm{Venus}} = \frac{W_{\mathrm{Venus,cRNAP}} + W_{\mathrm{Venus},\sigma_{70}} \, b_{\mathrm{Venus},\sigma_{70}}}{1 + W_{\mathrm{Venus,cRNAP}} + W_{\mathrm{Venus},\sigma_{70}} \, b_{\mathrm{Venus},\sigma_{70}} + W_{\mathrm{Venus,GntR}} \, b_{\mathrm{Venus,GntR}}} \tag{S20}$$

D-gluconate relieves repression by binding GntR and preventing it from occupying the operator. The effective GntR concentration available for repression is $[\mathrm{GntR}]_{\mathrm{eff}} = (1 - f_{\mathrm{bound}}) \, [\mathrm{GntR}]_{\mathrm{total}}$, where:

$$f_{\mathrm{bound}} = \frac{[\mathrm{gluconate}]^{n_{\mathrm{gluc}}}}{[\mathrm{gluconate}]^{n_{\mathrm{gluc}}} + K_{\mathrm{gluc}}^{n_{\mathrm{gluc}}}} \tag{S21}$$

is the fraction of GntR bound to D-gluconate (and therefore unable to repress), K_{gluc} is the dissociation constant, and n_{gluc} is the Hill coefficient for cooperative gluconate binding to the GntR dimer. At saturating gluconate ($f_{\mathrm{bound}} \to 1$), repression is fully relieved. Finally, since σ_{70} is supplied exogenously as protein and is not encoded on a gene in the circuit, its constitutive promoter control function is $\bar{u}_{\sigma_{70}} = W_{\sigma_{70},\mathrm{RNAP}}/(1 + W_{\sigma_{70},\mathrm{RNAP}})$ with $W_{\sigma_{70},\mathrm{RNAP}} = \exp(-0.5)$ fixed.

The partition function formalism provides a principled bridge between molecular-level binding events and macroscopic gene expression rates. Each pseudo-energy ϵ_s encodes the net free energy change for a specific promoter configuration relative to the empty state, integrating contributions from protein-DNA contacts, conformational changes, and steric effects. The resulting control function $\bar{u}_j$ is a smooth, bounded function that modulates the kinetic rate $r_{X,j}$ at each time step, coupling the thermodynamic state of the promoter to the dynamic mass balance equations.

S4. Consumable resource depletion

The second layer of the resource model describes the irreversible depletion of consumable resources (nucleotides, energy cofactors, amino acids) that cannot be regenerated in the cell-free system. We define $\epsilon_X(t) \in [0,1]$ and $\epsilon_L(t) \in [0,1]$ as the fractional availability of transcriptional and translational consumable resources, with $\epsilon_X(0) = \epsilon_L(0) = 1$. The depletion dynamics are:

$$\frac{d\epsilon_X}{dt} = -\alpha_X \left(\sum_{j=1}^{\mathcal{N}} l_{G,j}\, r_{X,j}\, \bar{u}_j \right) \epsilon_X \tag{S22}$$

$$\frac{d\epsilon_L}{dt} = -\alpha_L \left(\sum_{j=1}^{\mathcal{N}} l_{P,j}\, r_{L,j} \right) \epsilon_L \tag{S23}$$

where α_X and α_L are consumption rate constants (units: inverse flux), $l_{G,j}$ is the gene length (nt), and $l_{P,j}$ is the protein length (aa). The depletion rates are proportional to the total biosynthetic flux weighted by sequence length, reflecting the stoichiometric requirement that longer genes consume more nucleotides per transcript and longer proteins consume more amino acids per polypeptide. The resource fractions multiply the production terms in the mRNA and protein balance equations, providing a mechanistic link between cumulative biosynthetic load and the progressive decline of cell-free reaction productivity.

S5. Complete model equations

The full model consists of 11 ordinary differential equations. Gene concentrations are constant (linear DNA, no replication): $d\mathcal{G}_j/dt = 0$ for $j = 1, 2, 3$. The mRNA balances couple transcription (modulated by promoter control and resource availability) with first-order degradation:

$$\frac{dm_j}{dt} = r_{X,j}\, \bar{u}_j\, \epsilon_X - \theta_{m,j}\, m_j \qquad j = 1, 2, 3 \tag{S24}$$

where $\theta_{m,j} = \delta_{m,j} \ln 2/t_{1/2,m}$ is the effective degradation rate constant, $\delta_{m,j}$ is a gene-specific modifier, and $t_{1/2,m}$ is the characteristic mRNA half-life. The protein balances have an analogous structure:

$$\frac{dp_j}{dt} = r_{L,j}\, \epsilon_L - \theta_{p,j}\, p_j \qquad j = 1, 2, 3 \tag{S25}$$

where $\theta_{p,j} = \delta_{p,j} \ln 2/t_{1/2,p}$. Together with the resource depletion equations (Eqns. S22–S23), the algebraic expressions for transcription control (Eqns. S19–S20), machinery allocation (Eqns. S13–S14), and kinetic rates (Eqns. S9, S10), this defines a closed system of 11 ODEs with 25 estimated parameters.

References

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